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SOFTWARE REQUIREMENTS & SPECIFICATION

LEMON

Cloud Surge / Avanade

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1. Introduction

1.1 Purpose

This document defines the product requirements, system scope, constraints, and acceptance criteria for LEMON as agreed by Cloud Surge, Avanade and UCL.

LEMON is a workflow authoring and verification platform for converting domain knowledge into structured, executable decision workflows. It is intended to help non-programmer domain experts create, inspect, validate, preview, save, review, and export workflows without writing code.

1.2 Project Overview

LEMON provides:

- an AI-assisted workflow authoring experience through a conversational orchestrator
- a visual workflow editor for graph-based decision logic
- workflow validation before save, export, or execution
- preview execution against manually entered inputs
- export of workflows into machine-consumable formats
- a workflow library for private and shared workflow management

The system is especially suited to decision-support workflows in domains such as healthcare, administration, operations, and policy formalisation.

The project is an attempt to simplify the formalisation, storage, and sharing of medical workflows/flowcharts in a safe and quick manner; allowing doctors to create rich workflows that represent best practices and common methods.The value proposition is the speed that it allows information to be recorded at. Whereas previously doctors have had to explain a workflow to someone technical, they can now specify the program themselves without knowledge of programming. Essentially at its core LEMON is an AI-native flowchart builder, constructed with logical statements such that it is possible to build an interpreter of the workflow that can construct it into purely deterministic code.

1.3 Scope

LEMON includes:

- workflow creation from natural language and uploaded diagrams
- visual editing of graph-based workflows
- typed variables and typed outputs
- calculations, decisions, and subprocess references
- validation of workflow structure and references
- single-run preview execution using manually supplied inputs
- export to JSON, Python, and PNG
- workflow library management, including optional publication and peer review

LEMON does not process patient datasets directly. Batch execution, data storage, and production data processing are handled by downstream systems that consume LEMON's exported format, but those systems are outside of the scope of this project.

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1.4 Definitions, Acronyms and Abbreviations

Term	Definition
LEMON	The complete platform, including frontend, backend, orchestration, validation, execution, and export capabilities.
Workflow	A structured graph representing decision logic, control flow, variables, calculations, subprocesses, and outputs.
Orchestrator	The LLM-driven assistant that interprets user instructions and uses tools to build or edit workflows.
Tool	A discrete backend operation callable by the orchestrator, such as adding a node, validating a workflow, or listing workflows.
Validation	Automated structural and semantic checking of workflow correctness.
Preview Execution	Single-instance interpretation of a workflow against manually provided inputs, used by the author to verify correctness before export. Not for production data processing.
Formalisation	The process of converting informal clinical knowledge (images, verbal descriptions) into structured, machine-readable workflow definitions.
Variable	A typed piece of workflow data, either user-provided or derived during execution.
Derived Variable	A variable produced by a calculation node or subprocess execution.
Subprocess	A workflow node that invokes another workflow and uses its result as a derived value.
Shared Library	The public or peer-reviewed workflow catalogue visible to multiple users.

1.5 Intended audience

This document is intended for:

- project stakeholders
- Developers of the project (team 27)
- supervisors and advisors

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2. Stakeholders and Users

2.1 Stakeholder Register

Stakeholder	Role	Contact
Cloud Surge	Project Supervisor	ilyas@cloudsurge.uk
Avanade	Advisor and Facilitator	tarun.b.arora@avanade.com
UCL	Project Supervisor	yun.fu@ucl.ac.uk
Team 27	Developers	Axel Weernink, Axel.weernink.24@ucl.ac.uk Jeet thakwani, Jeet.Thakwani.24@ucl.ac.uk Jude Hawrani, Jude.hawrani.24@ucl.ac.uk James Mann, James.mann.24@ucl.ac.uk
Vallour	Clinical / Product Lead	Haider Khokhar, haider@triggerpointlabs.com

2.2 User Personas

The primary user will be any kind of domain expert who wants to formalise decision logic into a workflow without writing code, specifically a use case from our stakeholders is a doctor (GP) who wants to quickly assess patient data using a system they already have knowledge of, or one that they find on the platform and trust. They will have limited technical ability, but strong knowledge of the medical domain. They will interface with the system on the website, primarily through the orchestrator chat window.

3. Product goals

The product goals are:

1. Reduce the effort required to formalise expert decision logic.
2. Provide a safe authoring environment with strong validation before execution or export.
3. Support review, sharing, and reuse of high-value workflows.
4. Enable export to formats suitable for downstream engineering systems.

4. Functional Requirements

Must / Should / Could / Won't

4.1 Workflow Authoring

Covers the creation of workflows through the editor and orchestrator.

Req ID	Requirement Description	Priority
FR1-001	The system shall analyse uploaded workflow diagrams and reconstruct them into structured workflows in the editor.	**Must**

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FR1-002	The visual editor shall support the node types start, process, decision, calculation, subprocess, and end.	**Must**
FR1-003	Users shall be able to create and edit workflows through natural-language interaction with the orchestrator.	**Should**
FR1-004	Users shall be able to create and manage typed variables with supported types string, number, boolean, date, and enum.	**Must**
FR1-005	The orchestrator shall be able to register variables before creating nodes that depend on them.	**Must**
FR1-006	Decision nodes shall support type-appropriate comparators and true/false branching during execution.	**Must**
FR1-007	Calculation nodes shall support a documented set of mathematical operators and store results as derived variables.	**Must**
FR1-008	Users shall be able to create compound conditions combining multiple comparisons using logical AND/OR.	**Must**
FR1-009	Users shall be able to create subprocess nodes that call other workflows and map inputs and outputs between parent and child workflows.	**Must**
FR1-010	The orchestrator shall use workflow tools to create, update, inspect, validate, and save workflows.	**Must**
FR1-011	The system shall allow users to switch between multiple workflows for editing and review.	**Must**
FR1-012	The system shall allow import of workflow JSON into the editor.	**Should**
FR1-013	The orchestrator reasons about workflow correctness during creation, flagging potential issues and Matching its output to the source material as closely as possible.	**Must**

4.2 Verification and Execution

Covers the processing of data-sets against the workflows.

Req ID	Requirement Description	Priority
FR2-001	The system shall validate workflows against a structural rule set before save, export, or execution.	**Must**
FR2-002	Validation shall check, at minimum, start node constraints, reachability, edge references, complete decision branching,	**Must**

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	variable references, subprocess references, and output consistency.	
FR2-003	Users shall be able to execute a workflow against manually entered input values and inspect the final output, execution path, and final variable context.	**Must**
FR2-004	Preview execution shall support calculations, derived variables, subprocess execution, and cycle protection.	**Must**
FR2-005	The system shall expose step-level execution information for visual playback and debugging.	**Should**
FR2-006	Users shall be able to pause, resume, and stop stepped execution.	**Should**

4.3. Export and Interoperability

Req ID	Requirement Description	Priority
FR3-001	Users shall be able to export workflows as structured JSON including all data relating to the workflow such as nodes, edges, variables, outputs, and metadata.	**Must**
FR3-002	Users shall be able to export workflows as compiled Python code representing the workflow logic.	**Must**
FR3-003	Users shall be able to export the workflow canvas as a PNG image for documentation and sharing.	**Should**
FR3-004	Exported Python shall preserve control flow, decision logic, and calculation logic.	**Must**

4.4 Workflow Library

Req ID	Requirement Description	Priority
FR4-001	Users shall be able to save workflows to a personal library for later editing, verification, execution, and export.	**Must**
FR4-002	Users shall be able to browse their saved workflows through a library interface.	**Should**

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4.5 Conversational and Long-Running Task Support

Req ID	Requirement Description	Priority
FR5-001	The chat interface shall stream orchestrator responses incrementally where supported.	**Should**
FR5-002	Long-running orchestrator tasks shall support cancellation by the user.	**Must**
FR5-003	The system shall preserve enough conversation state to continue a workflow-building session across multiple turns.	**Must**
FR5-004	The system should allow a running task to resume or reconnect after client refresh where feasible.	**Should**

5. Non-Functional Requirements

5.1 Performance

The orchestrator relies on the Anthropic Claude API. Response times are bounded by API latency and the number of tool calls required.

Req ID	Requirement	Acceptance Criteria	Priority
NFR-001	Typical non-LLM UI actions should feel responsive.	Editor interactions, local state updates, library tab changes, and export initiation complete within 2 seconds on target hardware under normal load, excluding external API latency.	**Should**
NFR-002	Validation shall complete quickly for normal workflows.	Validation of typical author-sized workflows completes within 2 seconds under normal load.	**Should**
NFR-003	The system shall manage LLM context limits automatically.	Older conversation history is compacted before context exhaustion while preserving recent context and workflow continuity.	**Must**
NFR-004	Preview execution shall be bounded and protected against runaway logic.	Execution enforces a maximum step limit and detects subprocess cycles.	**Must**

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5.2 Reliability

Req ID	Requirement	Acceptance Criteria	Priority
NFR-005	Workflow persistence shall use a canonical save contract.	Nodes, edges, variables, outputs, output type, and tree remain synchronized after save/update operations.	**Must**
NFR-006	The system shall fail loudly for invalid workflow states rather than silently ignoring structural errors.	Invalid operations return explicit validation or execution errors.	**Should**

6. Data Requirements

6.1 Data Requirements

The system may store:

- workflow definitions
- workflow metadata
- variable and output definitions
- validation metrics
- workflow publication and voting metadata
- conversation and audit metadata
- uploaded workflow artifacts required for authoring continuity

6.2 Data Not in Scope for Long-Term Product Storage

The system shall not be used as a long-term store for:

- patient datasets
- bulk execution records over operational populations
- downstream clinical deployment outputs

6.3 Workflow data Lifecycle

Workflows are kept private by users and only stay in their account. They can also be deleted.

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7. Constraints, Assumptions, and Dependencies

7.1 Constraints

The system will be developed over 3 months by a team of 4 developers. The team will have access to cloud resources through Avanade which will allow a production MVP. Use of LLM APIs is costly, and could bump up against budget constraints if the system is heavily used. The product depends on external LLM inference services for orchestration, and LLM-backend features are subject to model latency, cost, and availability.

7.2 Assumptions and Dependencies

The project assumes steady access to the Anthropic API. It also assumes uptime from Azure, where the project is to be hosted.

8. Risks

Risk ID	Description	Likelihood	Impact	Mitigation
R-001	Sensitive operational or patient-identifiable data is entered into authoring workflows.	Low	Very High	Limit product scope to workflow authoring and preview, avoid dataset storage as a feature, document safe-use expectations, and enforce authenticated access.
R-002	LLM misuse, prompt injection, or malicious workflow instructions degrade safety or reliability.	Medium	High	Restrict orchestrator actions through explicit tools, validation, authenticated sessions, and server-side workflow checks.
R-003	Incorrectly authored workflows are mistaken for clinically approved production logic.	Medium	Very High	Require validation, preview execution, review flows, and downstream human review before any production deployment.
R-004	External API cost or latency impacts usability.	Medium	Medium	Use bounded orchestration flows, context compaction, and local validation/execution where possible.

9. Acceptance Criteria

The project will be accepted depending on the system behaving consistently with the scope and safety limitations defined in this document and fulfilment of each feature, requirement in this document unless alternatively agreed otherwise. Cloud Surge will be the primary point of contact for approval.

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10. Testing and Evaluation Strategy

10.1 Deterministic Testing

The testing strategy for LEMON shall combine automated and manual verification across the full product surface. Validation testing shall confirm that both strict and non-strict validation modes behave correctly, including type-safe conditions, branch completeness, reachability, and node and edge integrity. Tool and orchestrator testing shall verify that individual tools work in isolation, that the orchestrator can call them correctly in sequence, and that state remains synchronized across direct and MCP-based execution paths.

Execution testing shall confirm that preview execution produces the expected outputs for known inputs, that calculation operators behave correctly, that subprocess execution behaves predictably, and that bounded-execution protections such as cycle detection and step limits work as intended. Export testing shall verify that JSON exports preserve the workflow structure and metadata, that Python compilation produces usable code or clear warnings where compilation is incomplete, and that PNG export generates a correct representation of the visible workflow.

Integration testing shall cover the interaction between frontend and backend layers, including workflow loading, persistence, WebSocket chat flows, task cancellation and reconnect behavior, and library and peer-review features. Manual acceptance testing shall then verify realistic end-to-end scenarios such as authoring a workflow from scratch, reconstructing one from an uploaded diagram, saving and reopening it, validating it, previewing its execution, and exporting it in supported formats.

10.2 Evaluation

Because LEMON relies on an LLM orchestrator for diagram interpretation, workflow reconstruction, and natural-language workflow authoring, the system shall include a formal evaluation process for measuring model quality. This evaluation process shall be distinct from deterministic software testing and shall focus on the accuracy, consistency, and regression behavior of the orchestrator when generating or modifying workflows.

Model evaluation shall use benchmark tasks and reference workflows to compare generated outputs against expected structures and behaviors. These evaluations should consider, where appropriate, the correctness of node creation, edge topology, variable definitions, conditions, outputs, and execution behavior. The purpose of this process is not only to assess current quality, but also to provide a repeatable basis for improving prompts, tool schemas, orchestration logic, and workflow-construction strategies over time.

Changes to the orchestration layer, prompt design, tool definitions, or workflow extraction logic should be assessed against these evaluations to detect regressions and identify improvements. The evaluation framework should therefore be treated as part of the product-quality process for LEMON, particularly for AI-assisted workflow generation features.

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11. Glossary

Term	Definition
Node	A single step in a workflow. Types: start (entry point), process (action/label), decision (conditional branch), calculation (mathematical operation), subprocess (reference to another workflow), end (terminal output).
Edge	A directed connection between two nodes, representing flow of control. Decision edges carry true/false labels.
Input variable	A typed parameter (string, number, boolean, date, enum) provided by the user or upstream system when the workflow is executed.
Derived variable	A variable produced by a calculation node or subprocess output during execution. Not directly settable by the user.
Compound condition	A decision condition combining multiple comparisons with AND/OR operators.
Context compaction	Automatic summarisation of older conversation messages when the LLM context window approaches capacity.

12. Appendices

[Supporting material: wireframes, process flows, data models, meeting notes, etc.]

Document Sign-Off

By signing below, the undersigned confirm that this requirements specification has been reviewed and approved. Approval indicates agreement that the scope, requirements, constraints, and acceptance criteria described in this specification are an appropriate basis for implementation and assessment.

Name	Role	Signature	Date
Haider Khokhar	Clinical/Product Lead	Signed by: Haider Khokhar 982239896FB4467...	26/03/2026
Ilyas Khan	CEO & Co-founder	Signed by: Ilyas Khan 73F17A09AAEB41F...	26/03/2026

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